

ISSW 2026 · Saturday 3 October · 09:00–16:00 · Peak Room

autoATES v3.0

Mapping Connaught Creek with open-source tools

Follow along on a laptop, or sit and watch. Both tracks are first-class.

Two tracks, one timetable

Follow along

Run the notebook for the current module. Python 3.11 + QGIS. If a step fails, copy that folder's outputs_reference and continue.

Observe

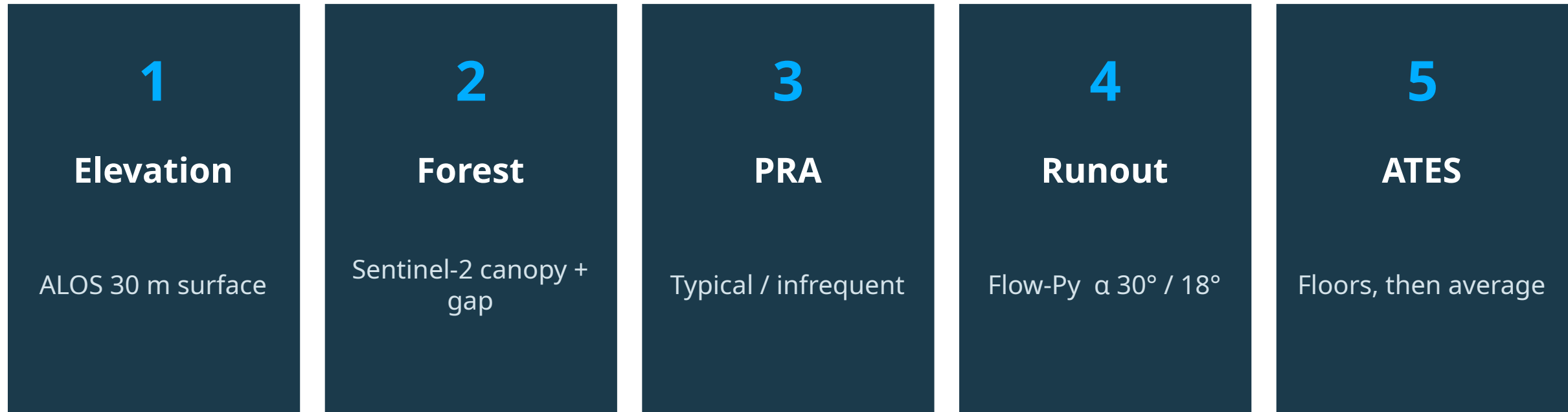
Same maps on this screen. No laptop required. You leave with the method, the design choices, and where to get the code.

Jump ahead

data/<step>/outputs_reference/ is the recovery path. Helpers will not stop the room for one laptop.

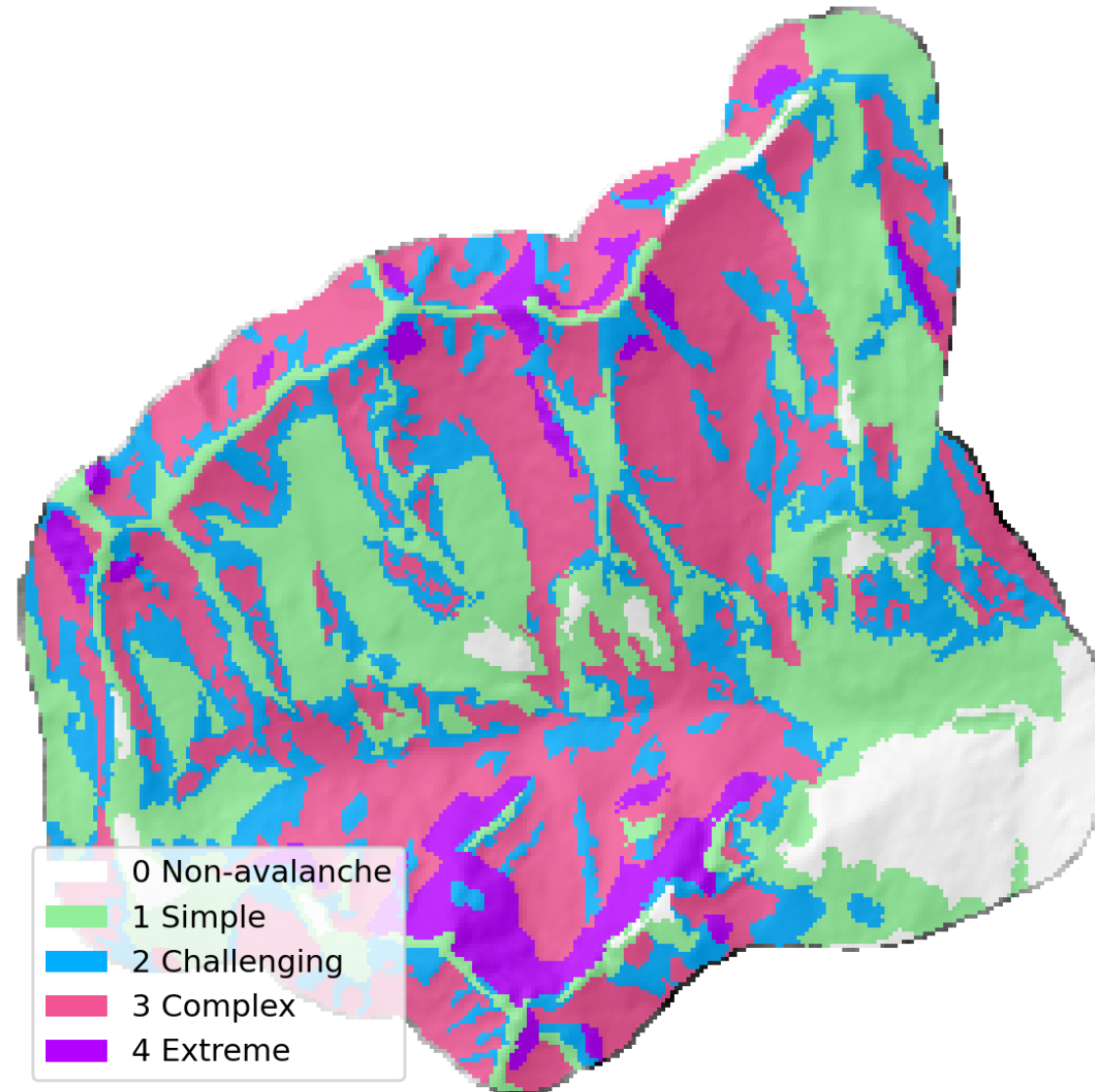
Helpers from 08:15. USB copies at the door. Conference wifi is not a plan.

Five steps. Typical and infrequent stay split.



One parameterization cannot stand in for everyday events and the larger, less frequent avalanches that still belong on an ATES map.

Connaught Creek, Rogers Pass — 13 km²

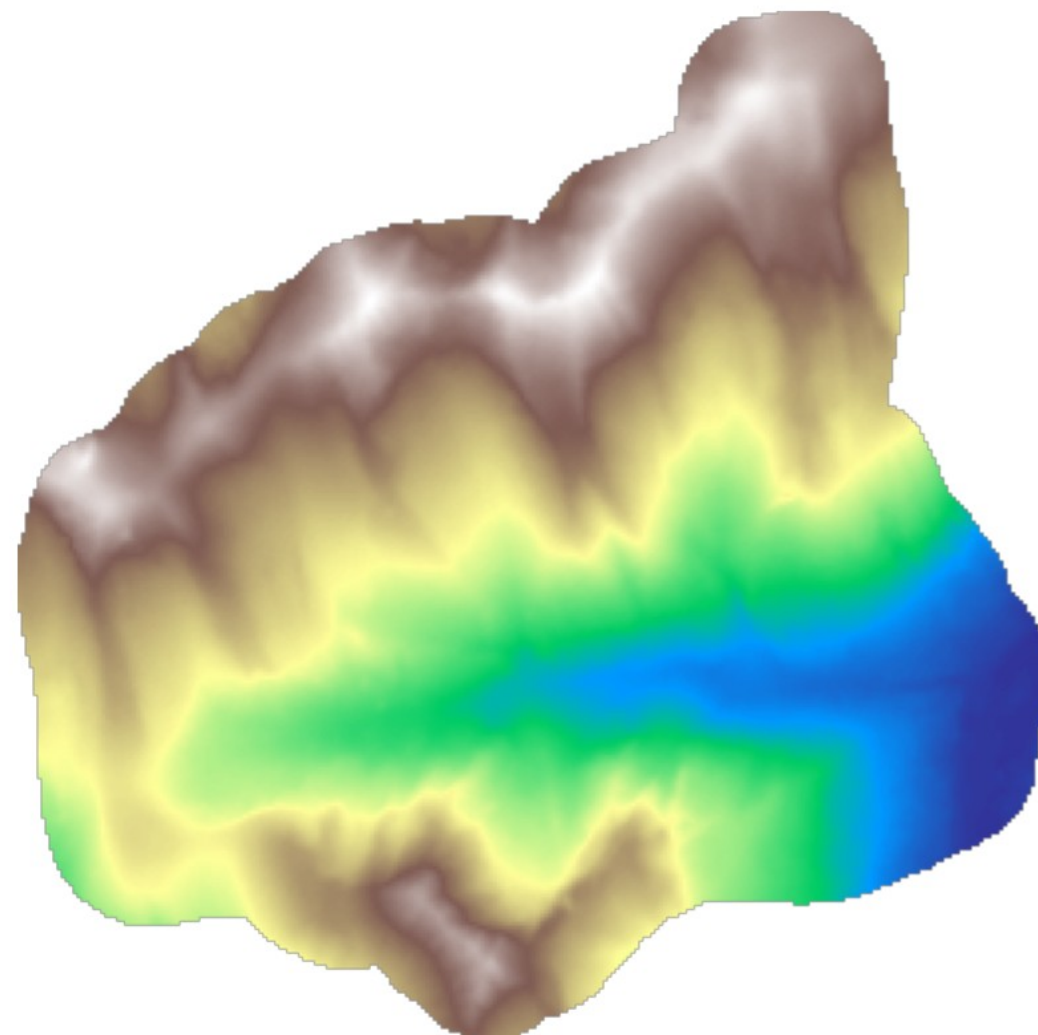


30 m surface. Canopy-top in forest. Slope is smoothed.

Hillshade



ALOS AW3D30 (m)



Summer greenness. Winter conifer, deciduous, and gully paths.

Summer NDVI



Winter NDSI



Binary conifer



PRA uses canopy cover. The classifier uses opening size.

Canopy cover % (3×3)

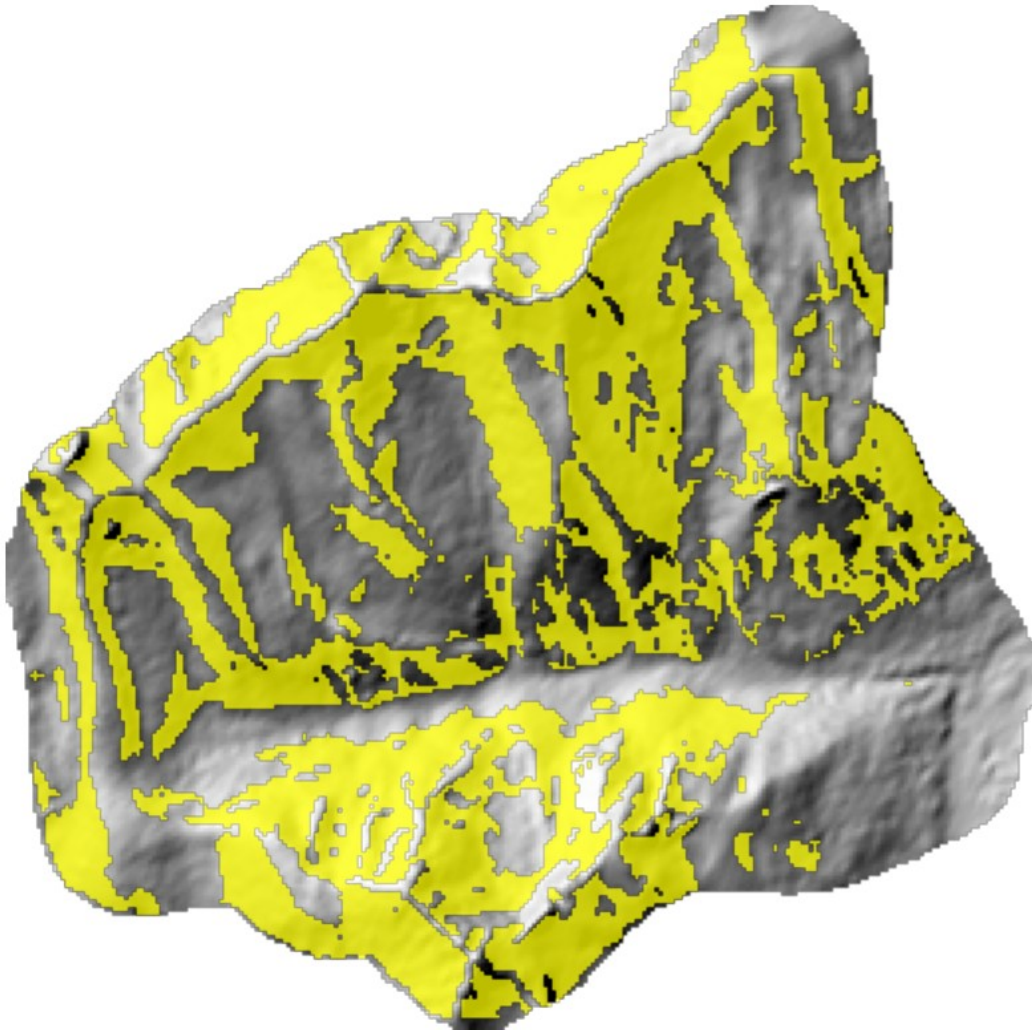


Gap area (ha)

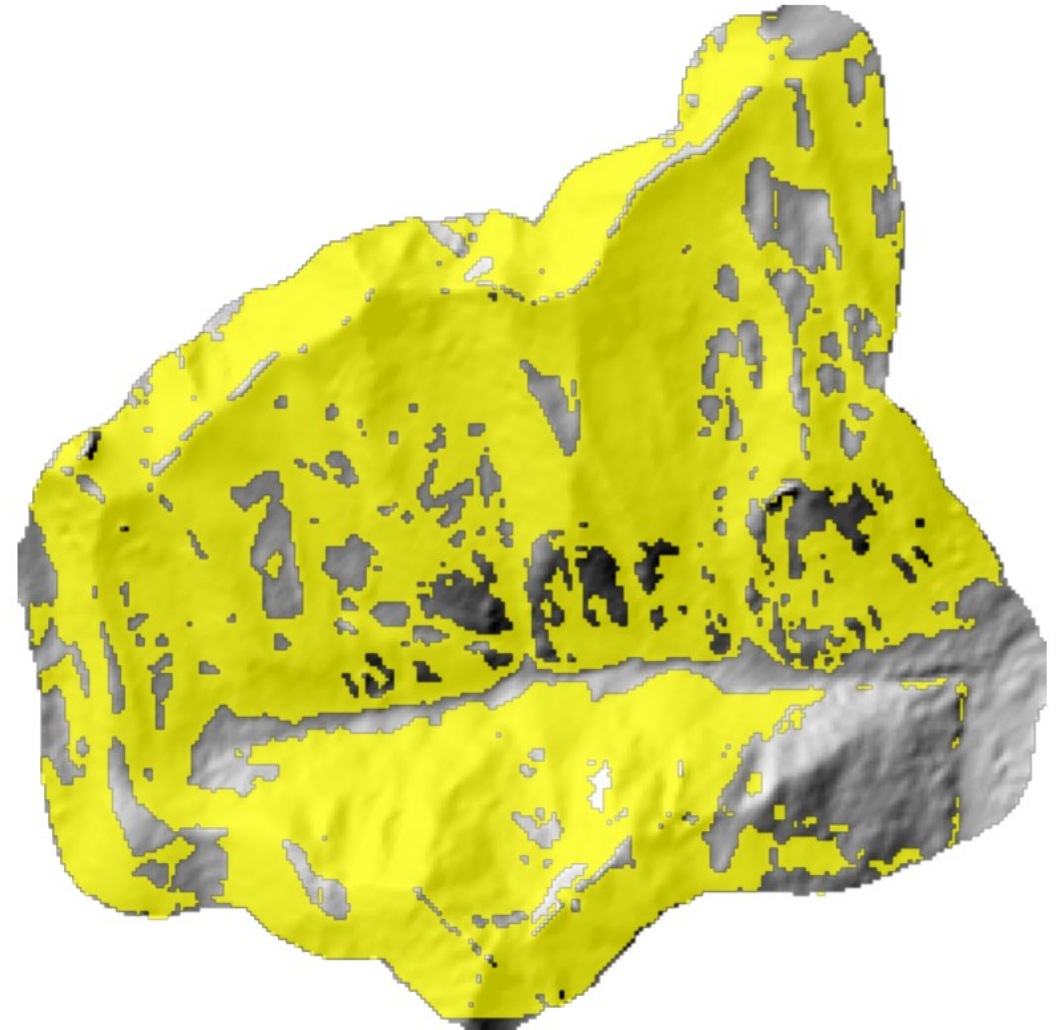


Typical is tighter. Infrequent picks up start zones below 30°.

Typical PRA

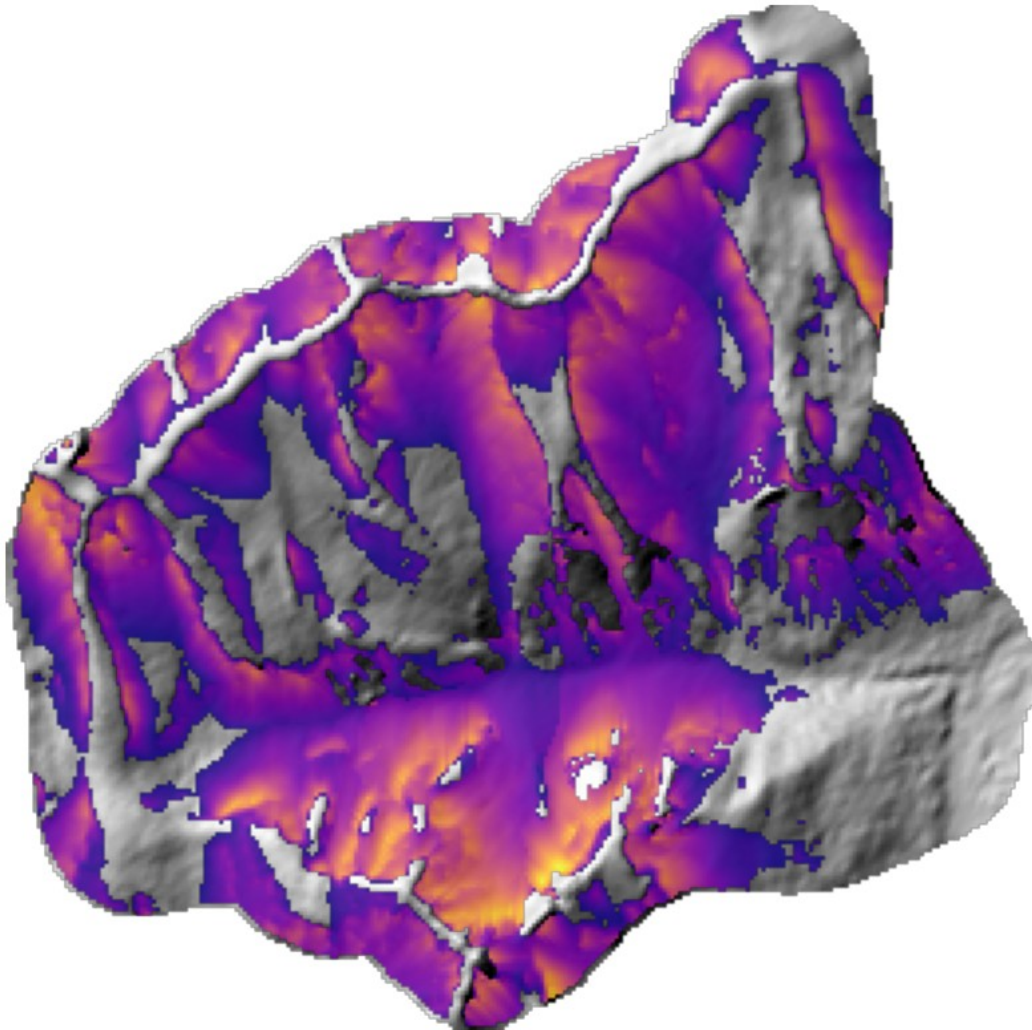


Infrequent PRA

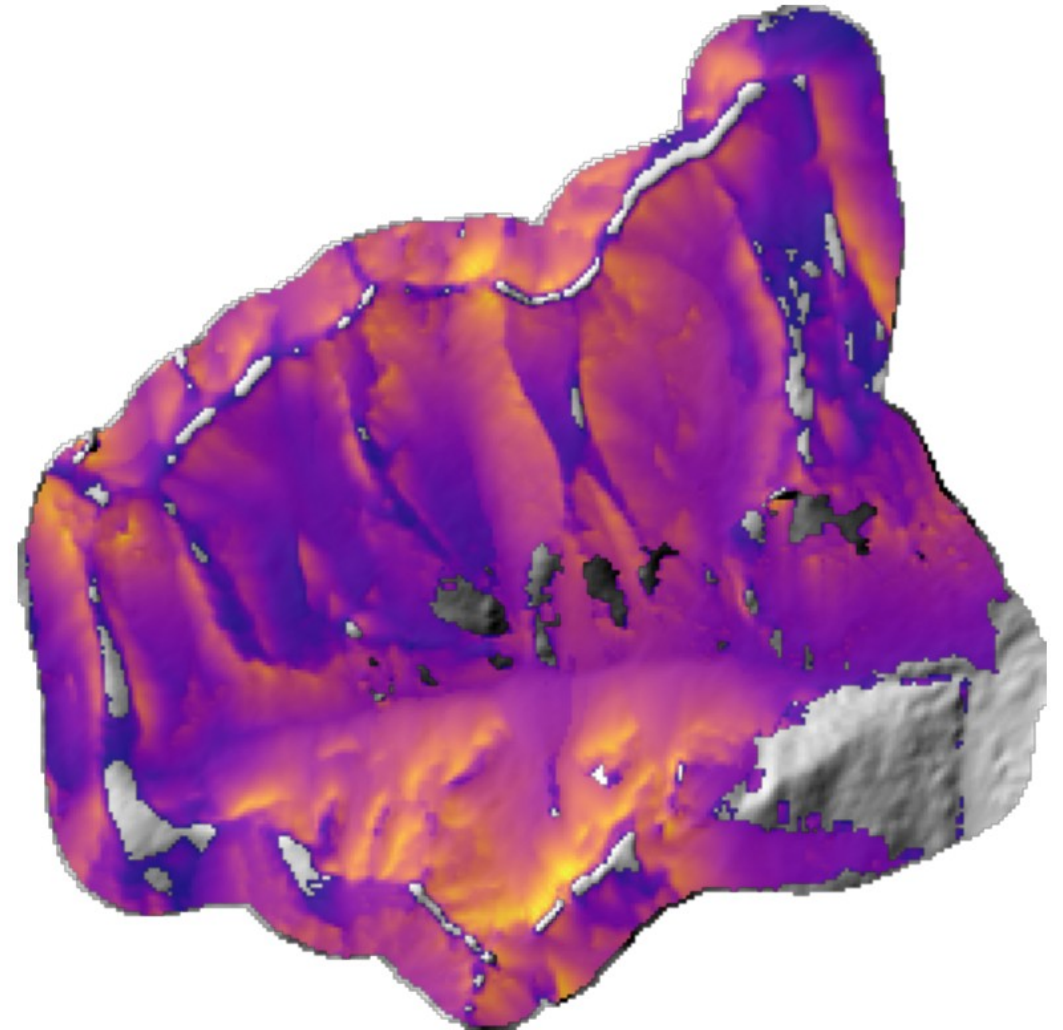


com4FlowPy — typical α 30°, infrequent α 18°. Not RAMMS.

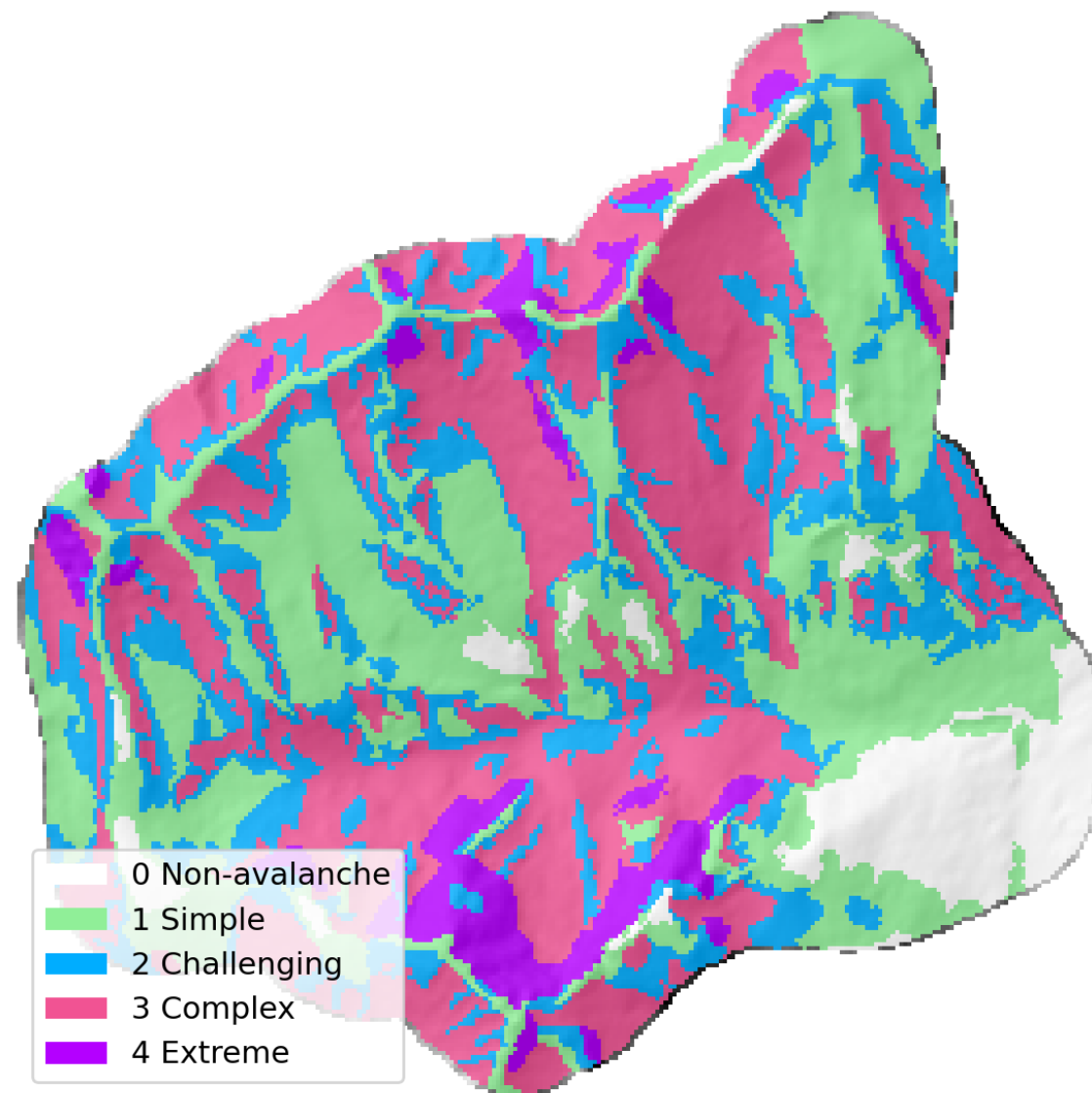
Typical reach · α 30°



Infrequent reach · α 18°



Infrequent sets the envelope. A floor cannot be averaged down.



Floors first. Then the rest is averaged.

Simple	Infrequent runout reach
Challenging	Typical runout reach
Complex	Typical overhead / contributing area
Extreme	Slope > 45° and open canopy

TAKE-HOME

Point this at a home range with an AOI, a 30 m surface, and a canopy raster.

Keep typical / infrequent parameters unless you have a reason and a validation set.

A map from this tutorial is not an operational ATES product.

Repo, pins, and papers: SETUP.md and CITING.md in the workshop repository.

Best-practices document is in development.